Note Title 9/8/2020

Unit 2.2 The limit of a function (p. 83–94)

Definition: If $f(x)$ is defined near the number a, then $\lim_{x\to a} f(x) = L$ is the limit of the function $f(x)$ as x approaches a.

We calculate limits by

① Direct substitution, e.g.

1. $\lim_{x\to 1}(x^2 - x + 4) = 1^2 - 1 + 4 = 4$

2. $\lim_{x\to 2} \frac{x^2-4}{x+2} = \frac{2^2-4}{2+2} = 0$

3. $\lim_{x\to \pi} \cos x = \cos \pi = -1$

② Factorization, e.g.

1. $\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(x+2)\cancel{(x-2)}}{\cancel{x-2}} = 4$

2. $\lim_{x\to 3} \frac{x^3-27}{x-3}$

ONE-SIDED limits

See e.g. the Heaviside function (p. 86)

Lefthandside limit: $\lim_{x\to a^-} f(x)$

Righthandside limit: $\lim_{x\to a^+} f(x)$.

NB: $\lim_{x\to a} f(x) = L \iff \lim_{x\to a^-} f(x) = L$ and $\lim_{x\to a^+} f(x) = L$. p.88

Examples:

1) $\lim_{x\to 1^-} \sqrt{x-1}$ does not exist, since if $x \to 1^-$, it means $x<1$ and $x-1<0$.

$\lim_{x\to 1^+} \sqrt{x-1} = 0$ $\left(\begin{array}{l} x\to 1^+ \\ \Rightarrow x>1,\ x-1>0 \end{array}\right)$

⑤ Consider $f(x) = \begin{cases} 2x+1 & \text{if } x<-2 \\ x^2-1 & \text{if } -2<x<1 \\ 1-x & \text{if } x>1 \end{cases}$.

Find (i) $\lim_{x\to 0} f(x)$ (ii) $\lim_{x\to 1} f(x)$ (iii) $\lim_{x\to -2} f(x)$

(i) $\lim_{x\to 0} f(x) = \lim_{x\to 0} x^2 - 1 = -1$.

(ii) To find $\lim_{x\to 1} f(x)$, we need to find $\lim_{x\to 1^-} f(x)$ and $\lim_{x\to 1^+} f(x)$.

$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (x^2-1) = \underline{0}$

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} 1-x = \underline{0}$

$\Rightarrow \lim_{x\to 1} f(x) = \underline{0}$

(iii) $\lim_{x\to -2^-} f(x) = \lim_{x\to -2^-} (2x+1) = \underline{-3}$

$\lim_{x\to -2^+} f(x) = \lim_{x\to -2^+} x^2-1 = \underline{3}$

$\Rightarrow \lim_{x\to -2} f(x)$ does not exist.

(3) Do example 4, p. 87.

Infinite limits: p. 89. (Read text book!)

See definitions 4 and 5 (p. 89, 90)

Examples:

(1) $\lim_{x\to 1} \frac{1}{(x-1)^2} = \infty$

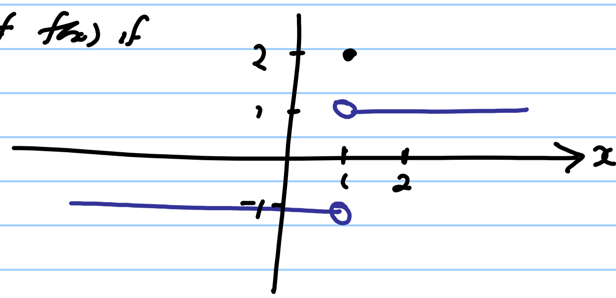
(2) $\lim_{x\to 1} \frac{4}{x-1}$. $\lim_{x\to 1^-} \frac{4}{x-1} = -\infty$, $\lim_{x\to 1^+} \frac{4}{x-1} = +\infty$ $\Rightarrow \lim_{x\to 1} \frac{4}{x-1}$ does not exist.

(3) Sketch the graph of $f(x)$ if

$\lim_{x\to 1^-} f(x) = -1$

$f(1) = 2$

$\lim_{x\to 1^+} f(x) = 1$



Vertical Asymptotes: Def 6, p. 90.

How to find a vertical asymptote:

Find the domain of f : D_f

If a is in D_f, then $x=a$ cannot be a vert. asymptote

If $a \notin D_f$, you need to TEST for a vertical asymptote:

Find $\lim_{x\to a^-} f(x)$ OR $\lim_{x\to a^+} f(x)$.

If one of these limits are at ∞ or $-\infty$, then $x=a$ is a vertical asymptote.

Do examples 7,8, p. 91.

Example:

Find the vertical asymptote of f if it exists:

(1) $f(x) = \frac{\sqrt{x}-3}{x+9}$

$D_f = [0,\infty) \Rightarrow$ No vertical asymptote.

② $f(x) = \frac{x-1}{x^2(x+2)}$

$\Rightarrow D_f = \mathbb{R} \setminus \{0, -2\}$

$\lim_{x\to 0} f(x) = \lim_{x \to 0} \frac{x-1}{x^2(x+2)} = \infty \Rightarrow x=0$ is a vert. asymptote

$\lim_{x\to -2^-} \frac{x-1}{x^2(x+2)}$ $\left(\frac{-3}{4(\text{small neg})}\right)$ $x \to -2^-$ $\Rightarrow x < -2$ $\Rightarrow x+2<0$

$= +\infty$

$\Rightarrow$ $\underline{x = -2}$ is a vertical asymptote.